

CLOUD RESOURCE USAGE ANOMALY DETECTION

A Machine Learning Approach to Identifying Abnormal Resource Consumption Patterns
in Cloud Infrastructure

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Project Report

Domain: Cloud Computing and Machine Learning

1. Introduction

Cloud computing platforms host a wide variety of workloads that continuously consume computing resources such as processing power, memory, storage input and output, and network bandwidth. Monitoring these resources is essential for maintaining system reliability, controlling operational cost, and preventing service disruption. When resource consumption deviates significantly from expected patterns, it is often referred to as an anomaly. Anomalies may indicate inefficient application behavior, unauthorized processes, or emerging system failures.

This report documents a project that applies exploratory data analysis and supervised machine learning to detect anomalies in cloud resource usage data. The analysis was carried out on a publicly available dataset that records minute level resource consumption metrics across multiple users and workload types, with each record labeled as either normal or anomalous.

2. Objectives

- Explore and understand the structure and characteristics of the cloud resource usage dataset
- Identify patterns and relationships between resource metrics and anomalous behavior
- Develop a machine learning model capable of classifying resource usage records as normal or anomalous
- Evaluate the model using standard classification metrics

3. Dataset Description

The dataset used in this project is titled Cloud Resource Usage Dataset for Anomaly Detection, obtained from Kaggle under a CC0-1.0 public license. It contains 14,400 records collected at one minute intervals, covering resource usage across ten distinct users and five workload types. Each record includes eight fields, described in Table 1.

Column	Description
Timestamp	Date and time of the recorded observation
CPU_Usage	Percentage of CPU utilization at the given timestamp
Memory_Usage	Percentage of memory utilization at the given timestamp
Disk_IO	Disk input and output activity
Network_IO	Network input and output activity
Workload_Type	Category of workload being executed
User_ID	Identifier of the user associated with the record
Anomaly_Label	Binary label, 1 for anomalous and 0 for normal

The dataset includes five workload categories: Database_Query, Video_Streaming, Web_Service, Crypto_Mining, and Backup. Of the 14,400 total records, 13,143 are labeled as normal and 1,257 are labeled as anomalous, representing an anomaly rate of approximately 8.7 percent.

4. Exploratory Data Analysis

4.1 Data Quality Assessment

An initial inspection confirmed the dataset contains no missing values across any of the eight columns. However, a small number of records contained negative values for CPU_Usage and Disk_IO, which are not physically meaningful for these metrics. Twenty three records had negative CPU_Usage values, all of which were labeled as normal. Similarly, 285 records had negative Disk_IO values, of which 252 were labeled as normal and 33 were labeled as anomalous. Since negative values did not correspond consistently with the anomaly label, they were treated as measurement noise rather than meaningful indicators of anomalous behavior.

4.2 Statistical Summary

Descriptive statistics showed an average CPU usage of approximately 35.6 percent, with values ranging up to 100.5 percent. Memory usage remained within a reasonable range of approximately 4 to 97 percent. Network_IO values were considerably smaller in magnitude compared to the other three metrics, ranging from approximately 0.5 to 42.4.

4.3 Temporal Pattern of CPU Usage

Figure 1 shows CPU usage over a single day for one user, with anomalous records highlighted. Normal usage fluctuated mostly between 0 and 50 percent, while nearly all anomalies corresponded to sharp spikes in the range of 80 to 96 percent. These spikes occurred consistently throughout the day rather than being concentrated in a specific time window.

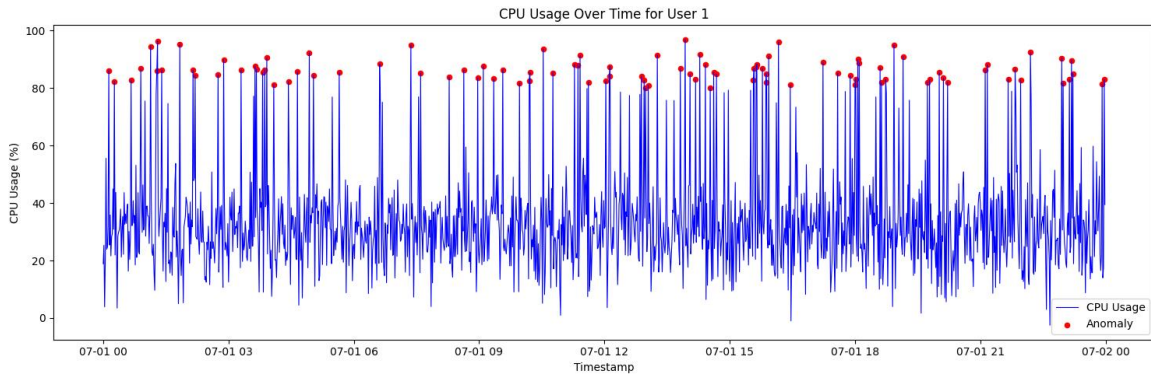


Figure 1: CPU usage over time for a single user, with anomalies marked in red

4.4 Feature Comparison Between Normal and Anomalous Records

The average values of each resource metric were compared between normal and anomalous records, as shown in Table 2. CPU usage increased sharply during anomalies, from an average of 30.7 percent in normal records to 86.3 percent in anomalous records. In contrast, memory usage, disk input and output, and network input and output showed only minor differences between the two classes.

Metric	Normal (Label 0)	Anomalous (Label 1)
CPU_Usage	30.71	86.29
Memory_Usage	44.55	39.99

Disk_IO	12.86	9.87
Network_IO	8.28	5.05

4.5 Anomaly Distribution Across Workload Types

Anomalies were found to be strongly associated with a single workload type. Table 3 summarizes the anomaly count and rate for each workload category. Four of the five workload types, namely Backup, Database_Query, Video_Streaming, and Web_Service, contained no anomalous records at all. The Crypto_Mining workload accounted for all 1,257 anomalies in the dataset, corresponding to an anomaly rate of 85.7 percent within that category.

Workload Type	Anomaly Count	Total Records	Anomaly Rate
Backup	0	1457	0.0 percent
Crypto_Mining	1257	1466	85.7 percent
Database_Query	0	4296	0.0 percent
Video_Streaming	0	2870	0.0 percent
Web_Service	0	4311	0.0 percent

Figure 2 presents a boxplot comparing CPU usage distributions across workload types. The plot visually confirms that Crypto_Mining workloads operate at a distinctly higher CPU usage range, between approximately 70 and 100 percent, with no overlap with the other four workload types.

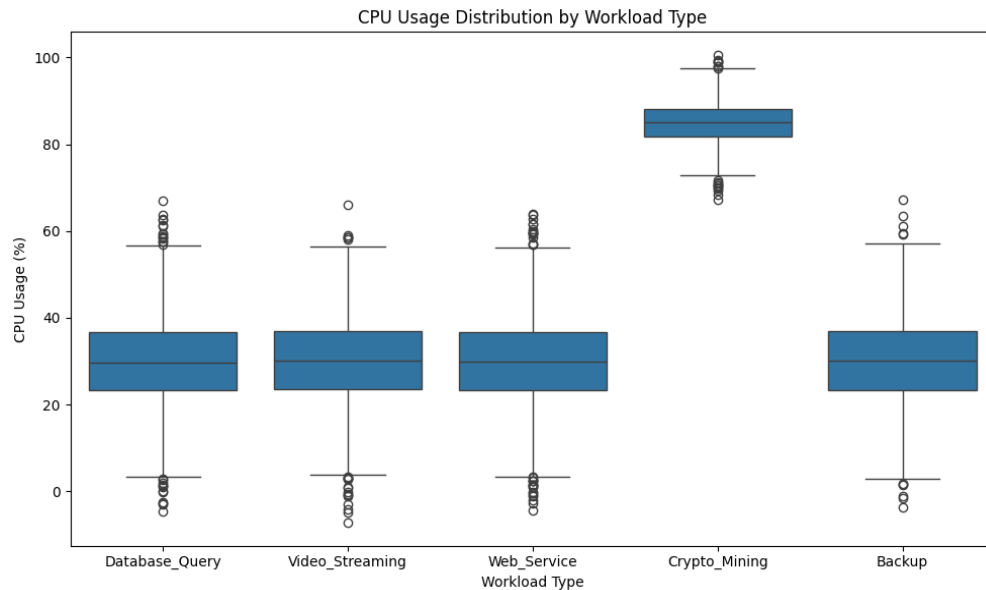


Figure 2: CPU usage distribution by workload type

4.6 Correlation Analysis

A correlation analysis was performed between the four numeric resource metrics and the anomaly label, presented in Figure 3. CPU_Usage exhibited a strong positive correlation of 0.82 with the anomaly label, confirming its role as the

primary indicator of anomalous behavior. Memory_Usage and Disk_IO showed a moderate correlation of 0.63 with each other but weak correlation with anomalies. Network_IO showed a weak negative correlation of negative 0.14 with anomalies.

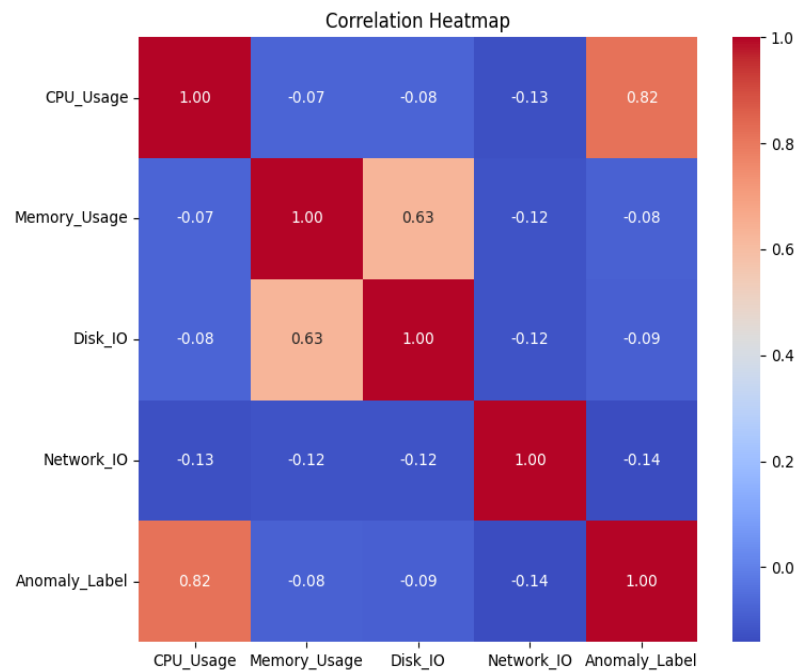


Figure 3: Correlation heatmap of resource usage metrics and anomaly label

5. Model Development

5.1 Feature Selection and Data Preparation

Based on the exploratory findings, five features were selected for model training: CPU_Usage, Memory_Usage, Disk_IO, Network_IO, and an encoded representation of Workload_Type. The Workload_Type column, being categorical, was converted into a numeric format using label encoding. The dataset was then divided into a training set and a testing set using an 80 to 20 split, with stratification applied to preserve the original proportion of normal and anomalous records in both subsets. The resulting training set contained 11,520 records and the testing set contained 2,880 records.

5.2 Model Selection and Training

A Random Forest classifier was selected for this task due to its robustness to noisy data, its ability to model non-linear relationships, and its capacity to report feature importance. The model was trained using 100 decision trees on the prepared training set.

6. Model Evaluation

6.1 Classification Performance

The trained model was evaluated on the held out test set of 2,880 records. The model achieved an accuracy of 100 percent, with precision, recall, and f1-score of 1.00 for both the normal and anomalous classes, as summarized in Table 4.

Class	Precision	Recall	F1-Score	Support
Normal (0)	1.00	1.00	1.00	2629
Anomaly (1)	1.00	1.00	1.00	251

This level of performance, while unusually high, is consistent with the exploratory findings, which showed that anomalies in this dataset are almost entirely determined by a combination of the Crypto_Mining workload type and sustained high CPU usage. This combination creates a clearly separable pattern between the two classes, which a Random Forest model is well suited to capture. In a dataset with more subtle or overlapping anomaly patterns, this level of accuracy would be less likely and would warrant further investigation.

6.2 Confusion Matrix

Figure 4 presents the confusion matrix for the test set. All 2,629 normal records were correctly classified as normal, and all 251 anomalous records were correctly classified as anomalous, with no false positives or false negatives observed.

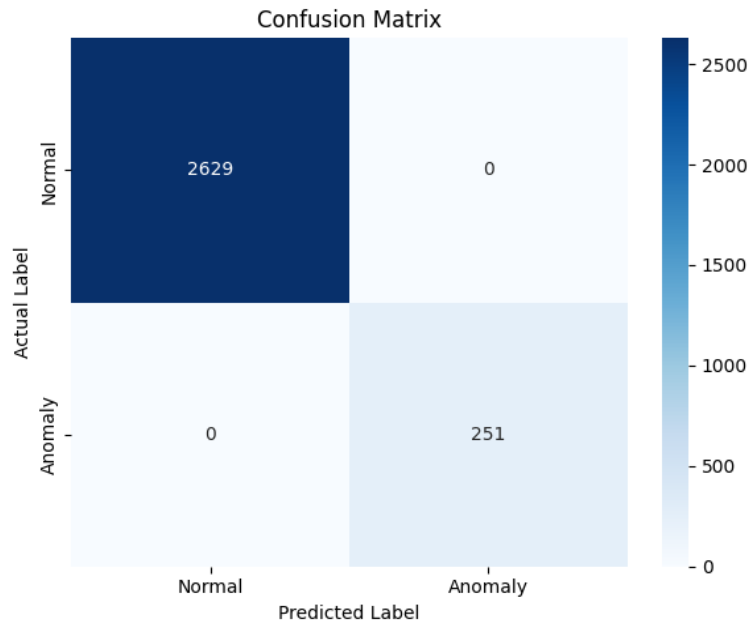


Figure 4: Confusion matrix on the test set

6.3 Feature Importance

Figure 5 shows the relative importance of each feature as determined by the trained Random Forest model. CPU_Usage was the most influential feature, accounting for approximately 67 percent of the model's decision making, followed by Workload_Type at approximately 28 percent. Together, these two features accounted for over 95 percent of the model's predictive power, while Memory_Usage, Disk_IO, and Network_IO contributed marginally. This outcome is consistent with the patterns identified during exploratory analysis, indicating that the model learned meaningful relationships rather than relying on noise.

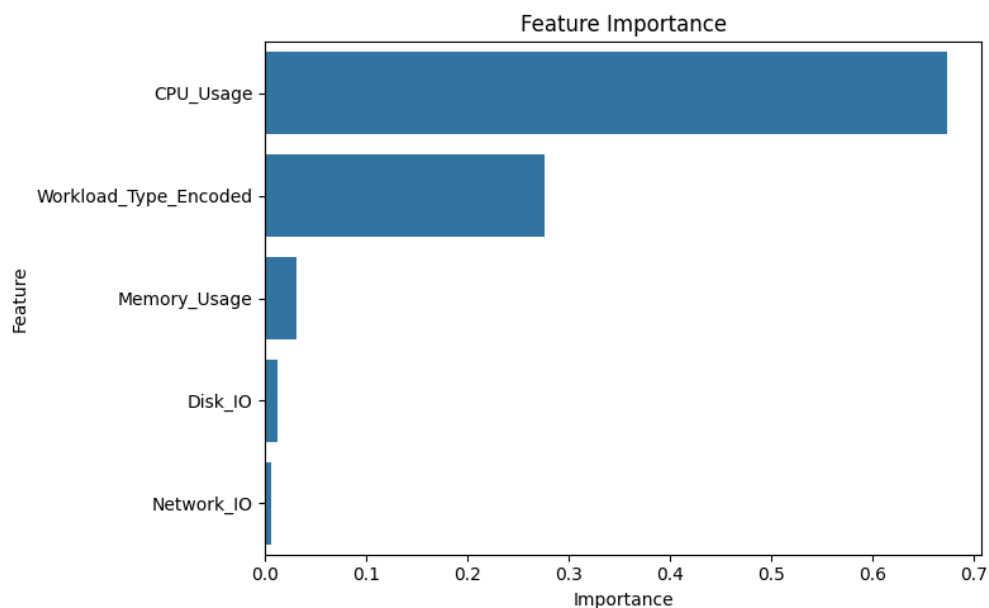


Figure 5: Feature importance from the Random Forest model

7. Discussion

The analysis conducted in this project indicates that anomalous resource usage in this dataset is closely tied to a specific workload category, `Crypto_Mining`, which is characterized by sustained high CPU consumption. This finding aligns with real-world expectations, as cryptocurrency mining processes are known to be computationally intensive and can produce resource usage patterns distinct from typical application workloads such as web services or database queries.

The Random Forest model demonstrated strong capability in identifying this pattern, achieving perfect classification performance on the test set. This result should be interpreted with an understanding of the underlying data structure. Because the anomalous class is strongly separable from the normal class based primarily on CPU usage and workload type, the classification task in this particular dataset is comparatively straightforward. In production cloud environments, anomalies may arise from a broader range of causes and may not exhibit such clearly separable characteristics, which would present a more challenging modeling scenario.

8. Project Summary

This project analyzed a cloud resource usage dataset containing 14,400 minute level records across five workload types and ten users, with the goal of detecting anomalous resource usage patterns. Exploratory analysis revealed that anomalies are almost entirely associated with the `Crypto_Mining` workload type, which showed sustained CPU usage between 70 and 100 percent, clearly separate from the 20 to 50 percent range observed in all other workload types. Other metrics, including memory usage, disk input and output, and network input and output, showed minimal difference between normal and anomalous records.

A Random Forest classifier was trained using `CPU_Usage`, `Memory_Usage`, `Disk_IO`, `Network_IO`, and `Workload_Type` as input features. The model achieved 100 percent accuracy, precision, recall, and f1-score on the test set, with no misclassifications. Feature importance analysis confirmed that `CPU_Usage` and `Workload_Type` together accounted for over 95 percent of the model's decision making, aligning with the patterns identified during exploratory analysis. This project demonstrates a complete and reproducible workflow for cloud resource anomaly detection, encompassing data exploration, statistical analysis, visualization, and machine learning model development.

9. Tools and Technologies Used

- Python as the primary programming language
- Pandas and NumPy for data manipulation
- Matplotlib and Seaborn for data visualization
- Scikit-learn for model development and evaluation
- Google Colab as the development environment

10. Dataset Source

Cloud Resource Usage Dataset for Anomaly Detection, Kaggle. Available at: <https://www.kaggle.com/datasets/programmer3/cloud-resource-usage-dataset-for-anomaly-detection>